VICTORY 2026

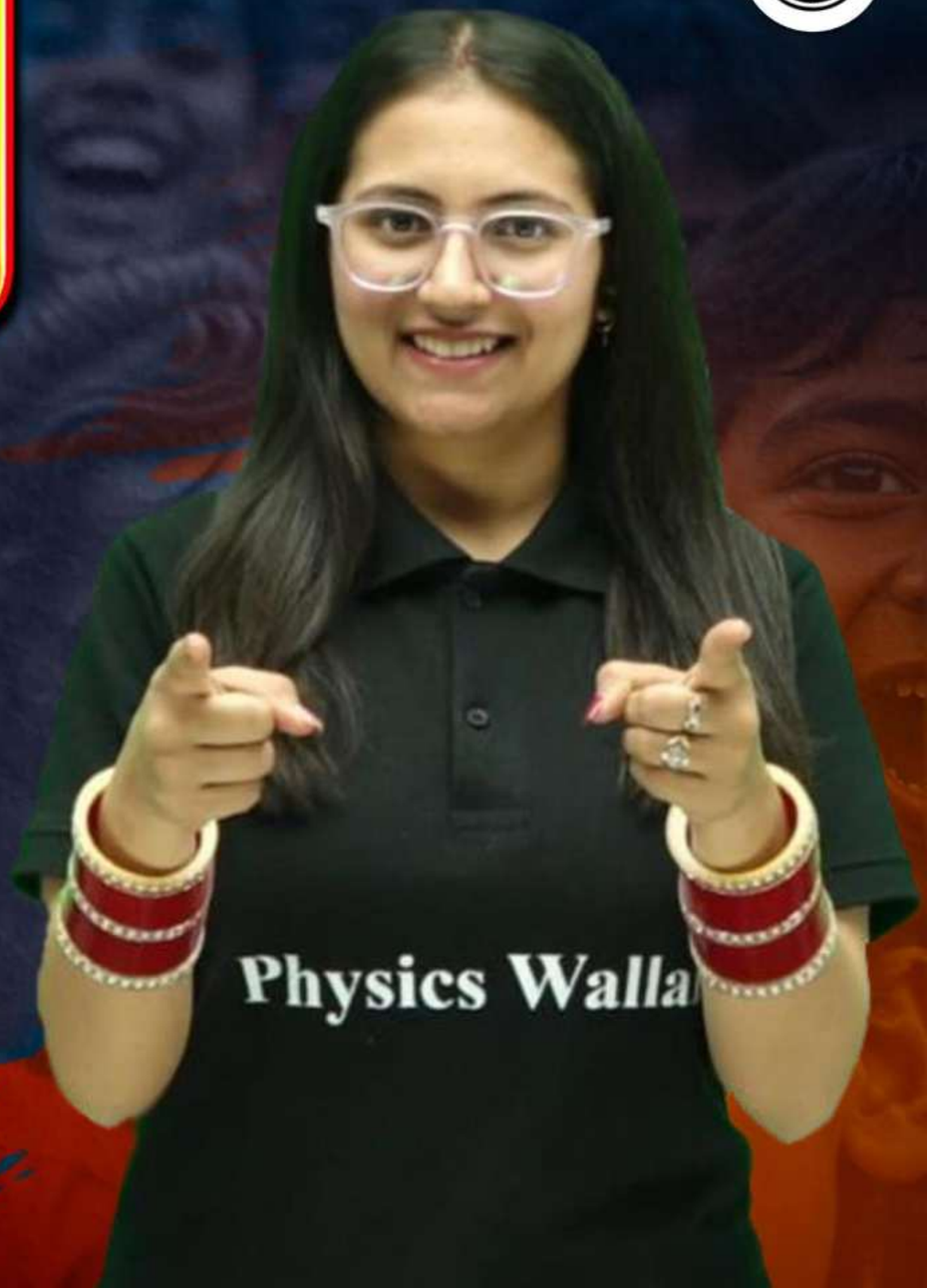
CHEMISTRY

Study of Compounds

Sulphuric Acid : H_2SO_4

Lect.-05

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Topics *to be covered*

A Sulphuric Acid in one-shot

B

C

D

Sulphuric Acid

H_2SO_4 ⇒ 'King of chemical'

* Non-Volatile acid
* Dibasic acid
* Comp. ionisation
* H_2SO_4 (Conc.)
  ↳ act as strong dehydrating Agent

H_2SO_4 :→ Strong mineral acid

→ show comp. ionisation

$$H_2SO_4 \rightleftharpoons 2H^+ + SO_4^{2-}$$

per acid Molecule

$2H^+ / 2H_3O^+$ ions

↳ 'dibasic acid'

Preparation of sulphuric acid

1) By oxidation of an aqueous solution of sulphur dioxide with oxygen, chlorine or bromine.

(1.) $2SO_2 + 2H_2O + O_2 \rightarrow 2H_2SO_4$

1.) $SO_2 + O_2 \longrightarrow SO_3$

2.) $SO_3 + O_2 \longrightarrow H_2SO_4$

(2.) $SO_2 + 2H_2O + Cl_2 \rightarrow H_2SO_4 + 2HCl$

Greenish yellow colour of chlorine fades.

(3.) $SO_2 + 2H_2O + Br_2 \rightarrow H_2SO_4 + 2HBr$

Reddish brown colour of bromine fades.

2) Sulphur with conc. nitric acid

$S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O.$

'HNO_3 act as good oxidising Agent'

Manufacture of sulphuric acid (contact process)

Amp

1. **Production of sulphur dioxide:** Sulphur dioxide is produced by roasting metallic sulphides in air

$4FeS_2$ (iron pyrites) + $11O_2$ → $2Fe_2O_3$ + $8SO_2$.

excess of oxy.

Burning in excess amount of oxy.

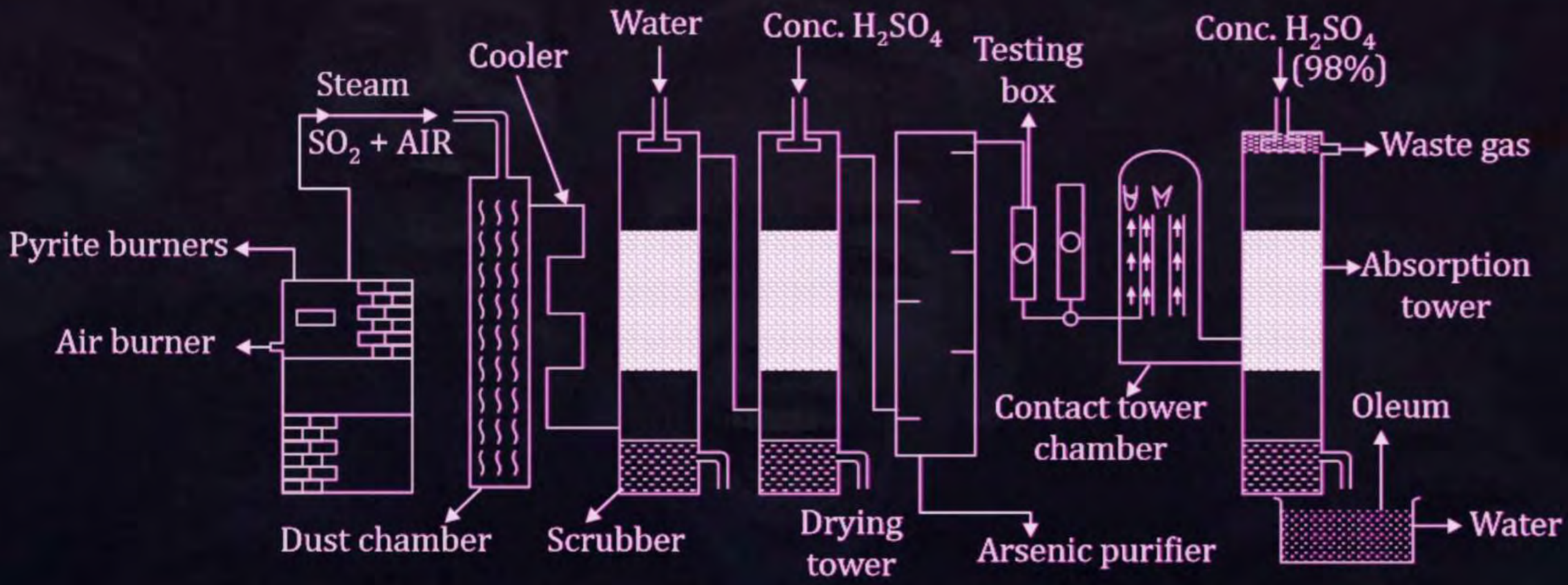
OR

by burning Louisiana sulphur (99·5% pure) in purified air.

$S + O_2 \rightarrow SO_2$.

2. **Purification of gases:** The mixture of sulphur dioxide and air contains various impurities which must be removed, otherwise the catalyst loses its efficiency.

Manufacture of sulphuric acid (contact process)



Manufacture of sulphuric acid (contact process)

iii) Catalytic oxidation of sulphur dioxide:

$$2SO_2 + O_2 \underset{450°C}{\overset{V_2O_5/pt}{\rightleftharpoons}} 2SO_3 + \Delta$$

$SO_2 + O_2 \underset{450^\circ c}{\overset{V_2O_5/Pt}{\rightleftharpoons}} 2SO_3 + Heat$

iv) Absorption of sulphur trioxide in sulphuric acid.

The gas is cooled in a heat exchanger and is then absorbed in absorption tower in concentrated sulphuric acid (98%).

$$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$$

Manufacture of sulphuric acid (contact process)

v) Dilution of oleum to obtain sulphuric acid.

It diluted in dilution tank by adding calculated amount of water to obtain sulphuric acid of the desired strength.

$$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$$



Conditions for the Conversion of SO_2 to SO_3

i) **Exothermic reactions are favoured at low temperature:**

The temperature should be as low as possible. The yield has been found to be maximum at about 410–450°C.



Conditions for the Conversion of SO_2 to SO_3

ii) High pressure:

High pressure favours the reaction because the product formed has less volume than reactant. But the acid-resistant towers which are able to withstand high pressure are difficult to build. Hence the pressure of 1 – 2 atmosphere is used.



Conditions for the Conversion of SO_2 to SO_3

iii) Excess of oxygen:

This increases the production of sulphur trioxide.

iv) A suitable catalyst:

Platinum is more efficient as a catalyst than vanadium pentoxide (V_2O_5), but platinum is more expensive and also it easily gets poisoned by impurities like arsenic (III) oxide.

Vanadium pentoxide is used as a catalyst.

Properties of Sulphuric Acid

Physical properties:

1.	Colour	Colourless.
2.	Odour	Odourless.
3.	Taste	Slightly sour in taste.
4.	Nature	Dense, oily, hygroscopic liquid. It absorbs moisture, so H_2SO_4 should always be kept stoppered.
5.	Density	Pure acid : 1.85 g/cc.
6.	Boiling point	338°C.
7.	Melting point	Pure sulphuric acid freezes to colourless crystals at 10.4°C.
8.	Solubility	Soluble in water in all proportions.
9.	Conductivity	Pure acid: Almost a non-conductor of electricity. Dilute acid: Good conductor of electricity.
10.	Constant boiling mixture	It forms a constant boiling mixture at 338°C containing 98.5% of the acid.
11.	Physiological nature	Concentrated acid is highly corrosive in nature and chars the skin black.

Chemical Properties

Pure (100 % H_2SO_4) hydrogen sulphate is not an acid as it does not ionize.

Sulphuric acid when dissolved in water form hydronium ion and so shows acidic properties.

amb

$$H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^-.$$

$$H_2SO_4^- + H_2O \rightleftharpoons H_3O^+ + SO_4^{2-}.$$

} Dibasic acid



Chemical Properties

$2H_3O^+$ ion

It ionizes in two stages and thus it is a dibasic acid.
It can form two types of salts, acid salt and normal salt.

$$NaCl + H_2SO_4 \xrightarrow{\text{below } 200^\circ C} NaHSO_4 + HCl$$
(Acid salt)

$$2NaCl + H_2SO_4 \xrightarrow{\text{above } 200^\circ C} Na_2SO_4 + 2HCl$$
(Normal salt)

The chemical properties of sulphuric acid depend on whether the acid is dilute or concentrated.

Properties of Dilute Sulphuric Acid

1. Acidic property:

Active metal	+	Acid (dil)	→	Metal Sulphate	+	Hydrogen gas ↑

Mg	+	H_2SO_4	→	$MgSO_4 + H_2\uparrow$
Zn	+	H_2SO_4	→	$ZnSO_4 + H_2\uparrow$
Fe	+	H_2SO_4	→	$FeSO_4 + H_2\uparrow$

Properties of Dilute Sulphuric Acid

2. It neutralises bases (metal oxides and metal hydroxides) to form salts and water.

Base/ Alkali + **Acid (dil)** → **Salt** + **Water**

$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$

$FeO + H_2SO_4 \rightarrow FeSO_4 + H_2O$

Metaloxide (Base)

Properties of Dilute Sulphuric Acid

3. It librerates carbon dioxide from metallic carbonates and bicarbonates.

Carbonate + Acid (dil) → Salt + Water + Carbon dioxide
bicarbonate

(Basic)

$Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2\uparrow$

$CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2\uparrow$



Properties of Dilute Sulphuric Acid

4. It evolves hydrogen sulphide from metal sulphides.

Metal Sulphide	+	Acid (dil)	→	Salt	+	Hydrogen Sulphide
Na_2S	+	H_2SO_4	→	Na_2SO_4	+	$H_2S\uparrow$
FeS	+	H_2SO_4	→	$FeSO_4$	+	$H_2S\uparrow$
ZnS	+	H_2SO_4	→	$ZnSO_4$	+	$H_2S\uparrow$

Properties of Dilute Sulphuric Acid

5. It evolves sulphure dioxide from sulphites and hydrogen sulphites (bisulphites)

Sulphite/ bisulphites	+	Acid (dil)	→	Salt	+	Water	+	sulphur dioxide
Na_2SO_3	+	H_2SO_4	→	Na_2SO_4	+	H_2O	+	$SO_2\uparrow$
$2NaHSO_3$	+	H_2SO_4	→	Na_2SO_4	+	H_2O	+	$SO_2\uparrow$

Properties of Concentrated Sulphuric Acid

1. Non-volatile nature:

Concentrated sulphuric acid has a high boiling point (338°C) and so, it is considered to be A non-volatile acid. It is, therefore, used for preparing volatile acids like hydrochloric-acid, Nitric acid and acetic acid from their salts by double decomposition.

Imp

Properties of Concentrated Sulphuric Acid

Salt of Volatile acid	+	Acid (conc.)	→	Acid salt	+	Volatile acid
$NaCl$	+	H_2SO_4	→	$NaHSO_4$	+	HCl
KCl	+	H_2SO_4	→	$KHSO_4$	+	HCl
$NaNO_3$	+	H_2SO_4	→	$NaHSO_4$	+	HNO_3
KNO_3	+	H_2SO_4	→	$KHSO_4$	+	HNO_3
CH_3COONa	+	H_2SO_4	→	$NaHSO_4$	+	CH_3COOH (Acetic Acid)
Sodium acetate		$H^+ + HSO_4^-$				

Properties of Concentrated Sulphuric Acid

2. As an oxidizing agent:

The oxidising property of concentrated sulphuric acid is due to the fact that on thermal decomposition, it yields nascent oxygen [O].

$$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$$

Properties of Concentrated Sulphuric Acid

(a) Non-metals

1. Carbon

$$C + 2H_2SO_4 \text{ (conc.)} \rightarrow CO_2 + 2H_2O + 2SO_2$$

2. Sulphur

$$S + 2H_2SO_4 \text{ (conc.)} \rightarrow 3SO_2 + 2H_2O$$

Properties of Concentrated Sulphuric Acid

(b) Metals

1. Copper

$$Cu + 2H_2SO_4\ (conc.) \rightarrow CuSO_4 + 2H_2O + SO_2 \uparrow$$

2. Zinc

$$Zn + 2H_2SO_4\ (conc.) \rightarrow ZnSO_4 + 2H_2O + SO_2 \uparrow$$

Imp

H_2SO_4 (Conc.)

Properties of Concentrated Sulphuric Acid

3. As a dehydrating agent

H_2SO_4 has a great affinity for water. It readily removes chemically combined water molecules i.e. elements of water from other compounds thus it acts as a dehydrating agent.

Properties of Concentrated Sulphuric Acid

(a) Conc. Sulphuric acid removes water of crystallization from ~~slats.~~ 'salts'

For example, blue copper (II) sulphate $CuSO_4.5H_2O$ taken in a test tube becomes (anhydrous copper sulphate), when a few drops of conc, sulphuric acid arc added.

$$\underset{\text{Blue}}{CuSO_4 \cdot 5H_2O} \xrightarrow{\text{Conc. } H_2SO_4} \underset{\text{Dirty white}}{CuSO_4(s)} + \underset{\text{(Removed by acid)}}{5H_2O}$$

Properties of Concentrated Sulphuric Acid

(b) All carbohydrates such as glucose, sugar cellulose (paper, cotton, wood, etc.) react immediately to give a black spongy mass of carbon which rises up. Steam is given off and the whole mass gets heated due to an exothermic reaction. They are said to get charred.

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Conc. } H_2SO_4} 6C + 6H_2O$$

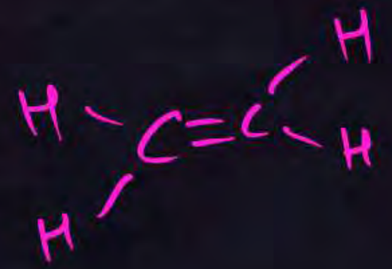
$$\underset{\text{Cane sugar}}{C_{12}H_{22}O_{11}(s)} \xrightarrow{\text{Conc. } H_2SO_4} \underset{\text{Sugar charcoal}}{12C(s)} + 11H_2O$$

Properties of Concentrated Sulphuric Acid

(c) Organic acids and organic compounds are dehydrated by conc. H_2SO_4 as follows:

$$\underset{\text{(Formic acid)}}{HCOOH} \xrightarrow{\text{Conc. } H_2SO_4} CO + H_2O$$

$$\underset{\text{(Oxalic acid)}}{(COOH)_2} \xrightarrow{\text{Conc. } H_2SO_4} CO + CO_2 + H_2O$$



$$\underset{\text{(Ethyl alcohol)}}{C_2H_5OH} \xrightarrow[170^\circ C]{\text{Conc. } H_2SO_4} \underset{\text{Ethylene}}{C_2H_4} \uparrow + H_2O$$

Properties of Concentrated Sulphuric Acid

4. Preparation of insoluble sulphates:

Sulphuric acid precipitates the insoluble sulphates of lead, barium and calcium from the solutions of their salts.

Soluble salt + Acid → Insoluble Salt + Acid

HNO_3

$Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4 + HNO_3$

$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$

HCl

Uses of Sulphuric Acid

1. It is used in the preparation of : a) Halogens, (b) Carbon monoxide, (c) Carbon dioxide, (d) Hydrogen, (e) Sulphur dioxide. Concentrated sulphuric acid is used to dry gases like HCl gas. **Note:** Gases like H_2S and hydrogen cannot be dried by it as they react. ($H_2 + H_2SO_4 \longrightarrow 2H_2O + SO_2$)	
2. Metallurgy i) Extraction	Its reaction with metallic compounds gives sulphates which on electrolysis give the metal in the pure form.
ii) Pickling metals	Removes metallic impurities like oxides and carbonates from the surface of metals before galvanizing.
3. In lead accumulators	It undergoes electrolysis in the aqueous state.
4. Oil refining	Sulphuric acid is used to remove harmful impurities in purification of oil products, e.g., petrol, kerosene and lubricants.

Uses of Sulphuric Acid

5. Industrial uses: in the manufacture of :

i) Fertilizers like
   (a) Ammonium sulphate [$(NH_4)_2SO_4$],
   (b) Superphosphate of lime [$Ca(H_2PO_4)_2 + CaSO_4$],

ii) Dyes, drugs
   (a) Artificial fibres: Rayon, Nylon.
   (b) Dyes, drugs [from coal tar derivatives].

iii) Explosives like
   (a) Tri-nitro toluene [T.N.T.],
   (b) Picric acid,
   (c) Tri-nitro glycerine [$C_3H_5O_3(NO_2)_3$].

Uses of Sulphuric Acid

5. Industrial uses: in the manufacture of :

iv) Acids like
- (a) Nitric acid [HNO_3],
- (b) Hydrochloric acid [HCLJ,
- (c) Phosphoric acid [H_3PO_4],
- (d) Acetic acid [CH_3COOH],

v) Compounds like
- (a) Sodium sulphate [glass industry],
- (b) Ferrous sulphate [ink industry].

Test for sulphuric acid and sulphates

1. When barium chloride solution is added to sulphuric acid or any soluble metal sulphate, white precipitate of barium sulphate is formed.

$$BaCl_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + 2HCl$$

$$BaCl_2 + MgSO_4 \rightarrow BaSO_4\downarrow + MgCl_2$$

The white precipitate of barium sulphate is insoluble in any mineral acid like hydrochloric acid or nitric acid.

Test for sulphuric acid and sulphates

2. Conc. Sulphuric acid on heating with copper evolves sulpur dioxide.

$$Cu + \underset{(Conc.)}{2H_2SO_4} \rightarrow CuSO_4 + 2H_2O + SO_2 \uparrow$$

Sulphur dioxide turns acidified dichromate solution green (Reducing property of SO_2). $K_2Cr_2O_7$ (orange)

$$\underset{(Orange)}{K_2Cr_2O_7} + H_2SO_4 + 3SO_2 \rightarrow \underset{(Green)}{Cr_2(SO_4)_3} + K_2SO_4 + H_2O$$

Test for sulphuric acid and sulphates

3. Conc., sulphuric acid on heating with NaCl evolves pungent fumes of HCl gas.

$$NaCl + \underset{(Conc.)}{H_2SO_4} \longrightarrow NaHSO_4 + HCl \uparrow$$

HCl gas gives dense white fumes of ammonium chloride with a glass rod dipped in ammonium hydroxide.

$$HCl + NH_4OH \rightleftharpoons \underset{(white)}{NH_4Cl}$$

Question



Which of the following metal will not react with dilute ~~sulphureic~~ H_2SO_4 (dil.) acid?

A Gold

B Iron

C Zinc

D Aluminium

Question



When metal reacts with dilute sulphuric acid, ... gas is produced.

A Oxygen ✗

B Sulphur dioxide

C Sulphure trioxide

D ✓ Hydrogen gas

Question



Which of the following compounds reacts with dilute sulphuric acid and produces a colourless gas with a rotten egg-like smell?

H_2S

A Metal hydroxide

B Metal carbonate

C Sugar

D Metal sulphide

Question



Which of the following will not ~~read~~ react with dilute suphuric acid

A Glucose

B Magnesium hydroxide

C Zinc sulphide

D Sodium carbonate

Question



What would happen if we dropped a few drops of concentrated sulphuric acid on paper?

A Nothing happens

B Paper will turn blue

C Paper will char

D Paper will oxidise

Question

The gas X evolved during the reaction shown in the picture is:



A O_2

B H_2

C SO_2

D CO_2

B



Question



Which of the following is not oxidised by concentrated sulphuric acid?

A Metals

B Diamond

C Non-metals

D Inorganic compounds

Question

Which of the following statement is false about dilute sulphuric acid?



A. It is a strong electrolyte.

B. It is a strong oxidising agent.

H.W

C. reacts with the metal to give hydrogen gas.

D. It ionises to a great extent.

Question



How many of the following compounds will give black spongy mass on reaction with concentrated sulphuric acid?
Glucose, cellulose, table salt, cane sugar, baking soda, fructose, starch, DNA

A 6

B 5

C 7

D 4

H.W

Question

The acid used in the reaction shown by the given picture is

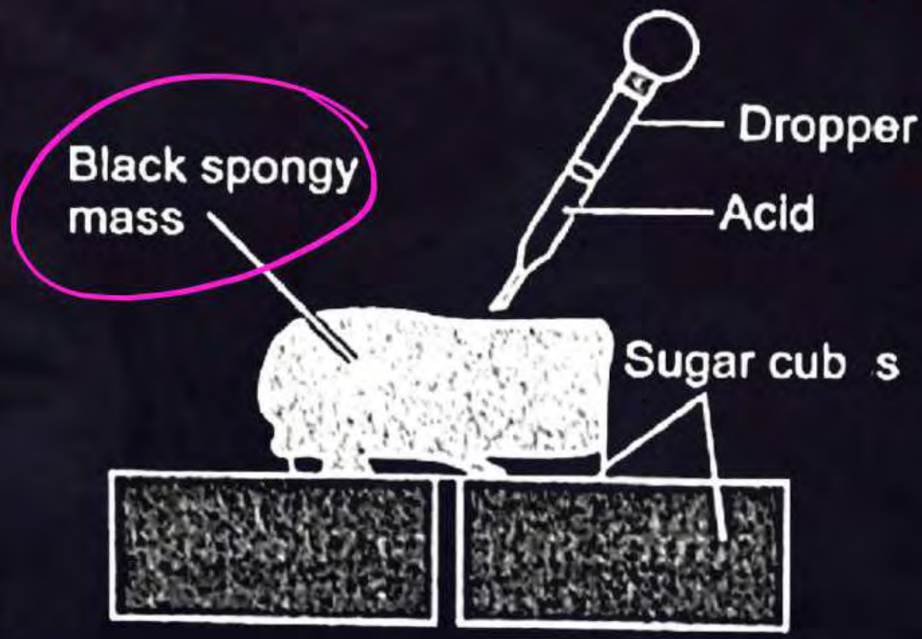


A Concentrated H_2SO_4

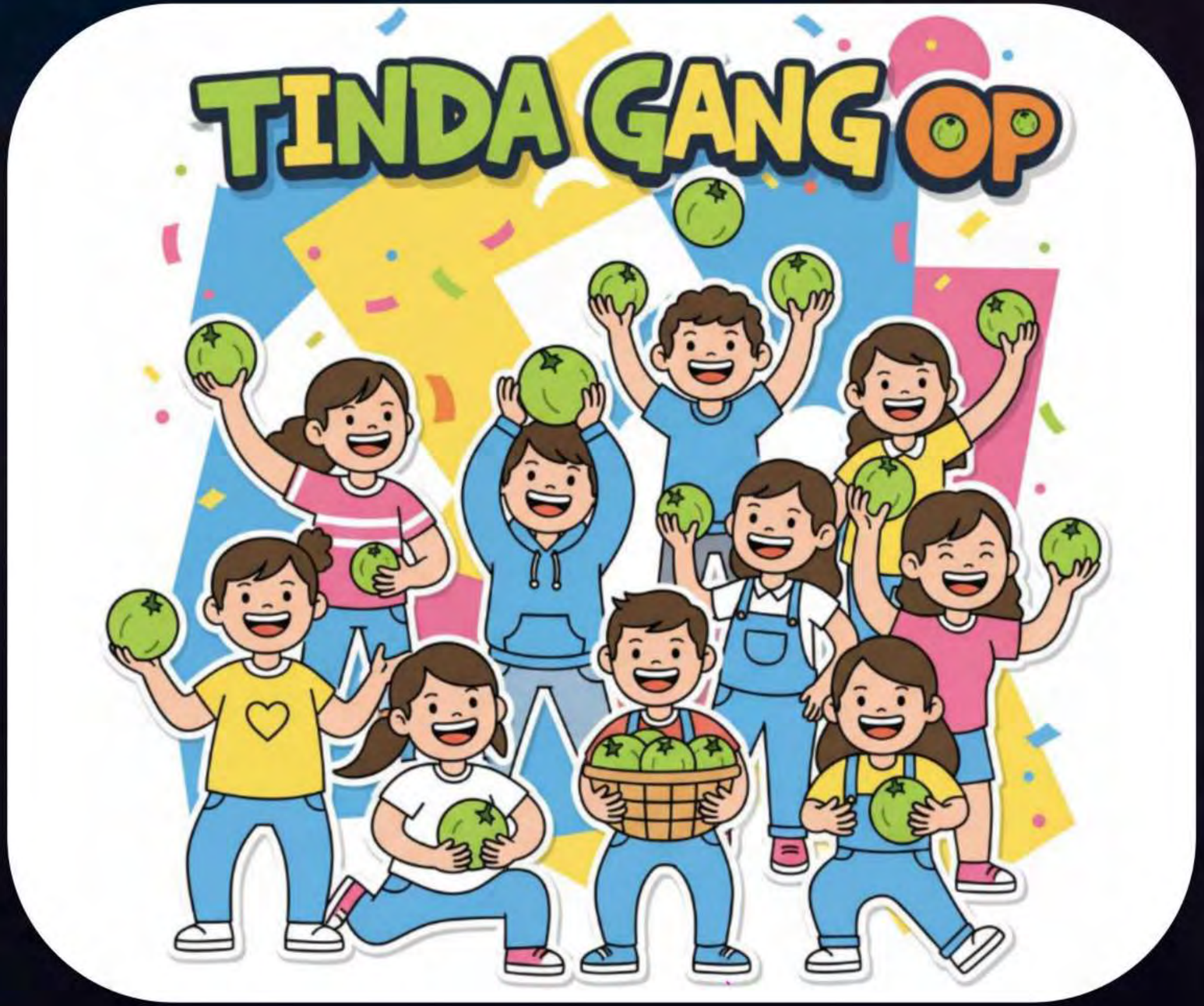
B Dilute H_2SO_4

C Acetic acid

D Citric acid




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95+
100

1.) Mole concept

2.) organic chemistry

3.) Metallurgy

4.) electrolysis

5.) Acid-Base

C.B.Q *

+

PYQ

'Paper pren' +

Thank-you



Good Health ∝